

Feature-Aligned Adaptive Triggers for Effective and Stealthy Clean-Label Backdoor Attacks

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Abstract

Clean-label backdoor attacks are appealing because they leave poison-label annotations untouched, yet current methods become unreliable when the victim task is hard—many-class, low-poison-rate, or high-base-accuracy regimes where a small, fixed trigger is diluted by the strong benign signal. We present **FAAT** (Feature-Aligned Adaptive Trigger), a clean-label attack composed of three generalized components: (i) a frozen global direction δ_g that supplies a strong, transferable backdoor signal; (ii) a bounded adaptive residual δ_a shaped in the DCT domain that restores effectiveness while staying imperceptible; and (iii) a feature-alignment loss $\mathcal{L}_{\text{align}}$ that pulls poisoned representations toward the target class so sample-clustering defenses cannot isolate them. FAAT is trained self-containedly, with no external pre-optimized noise. On CIFAR-10, CIFAR-100 and Tiny-ImageNet, FAAT raises attack success rate by +3.9 to +51.9 points over the strongest reproduced clean-label baseline, with benign accuracy intact (SSIM 0.92–0.98, $\ell_2 \leq 1.5$ on natural images), and is not flagged by Activation Clustering, Spectral Signatures, STRIP or Fine-Pruning. As a complementary contribution we introduce **KST**, a trigger whose spectrum is structurally flat (peak/median ≈ 1.0 , vs. 68.8 for Narcissus), defeating frequency-domain defenses at ASR parity. Finally, we treat GTSRB not as a fourth dataset but as a stress test of overconfident victims ($BA \approx 100\%$): there, fixed, flat-spectrum and adaptive triggers each reveal a distinct failure mode, exposing a fundamental ASR–stealth trade-off that tightens as victim confidence grows.

1. Introduction

Deep neural networks are known to be vulnerable to *backdoor attacks*: an adversary poisons a fraction of the training data so that the learned model behaves normally on clean inputs but misclassifies any input bearing a trigger to a target label [6]. Most early attacks *flip the label* of poisoned sam-

ples, which is easy to spot under visual inspection. **Clean-label** attacks [3, 25] are stealthier: poisoned samples keep their *true* (target-class) label, so the attack is invisible to label-level auditing. Their price is effectiveness: without a label signal telling the model *where* the trigger is, the model must discover the trigger on its own, which requires either large triggers or careful sample selection.

A recent line of work [30] systematizes this with three *generalized components*—a quantization/blend trigger, a sample-selection criterion, and a clean-label training protocol—and shows that the right combination (e.g. a residual-squared selection) recovers competitive attack success rates (ASR) on CIFAR-10. **However, the approach is brittle beyond that comfort zone.** We reproduce these components faithfully and observe two failure modes (Fig. 3): (1) on *many-class / low-poison* regimes (CIFAR-100 at 0.5%, Tiny-ImageNet at 0.25%) the strongest baseline reaches only $\sim 44\text{--}85\%$ ASR; (2) on *high-base-accuracy* tasks (GTSRB, $BA \approx 100\%$) the fixed trigger is diluted to $\leq 34\%$ ASR—and two of four baselines collapse to 0%. In all cases the benign task is so easily learned that the tiny poison signal is washed out.

We argue the fix is to make the trigger *adaptive and feature-aware* rather than fixed. We propose **FAAT** (Feature-Aligned Adaptive Trigger), which adds three components on top of the clean-label framework:

- A **frozen global direction** δ_g (Narcissus-style [33]) supplies a strong, transferable backdoor signal that is robust to benign dilution.
- A **bounded adaptive residual** δ_a , optimized *per poisoned image* in the DCT domain and clipped to a small ℓ_2 budget, restores attack effectiveness without sacrificing imperceptibility.
- A **feature-alignment loss** $\mathcal{L}_{\text{align}}$ pulls the representations of poisoned samples toward the target-class centroid, so that Activation-Clustering [2] and Spectral-Signature [24] defenses cannot separate them from clean target-class samples.

All three are trained *self-containedly*—FAAT does not de-

FAAT poisons target-class images with an imperceptible trigger ($L_2 \approx 1.5$, $SSIM > 0.95$)

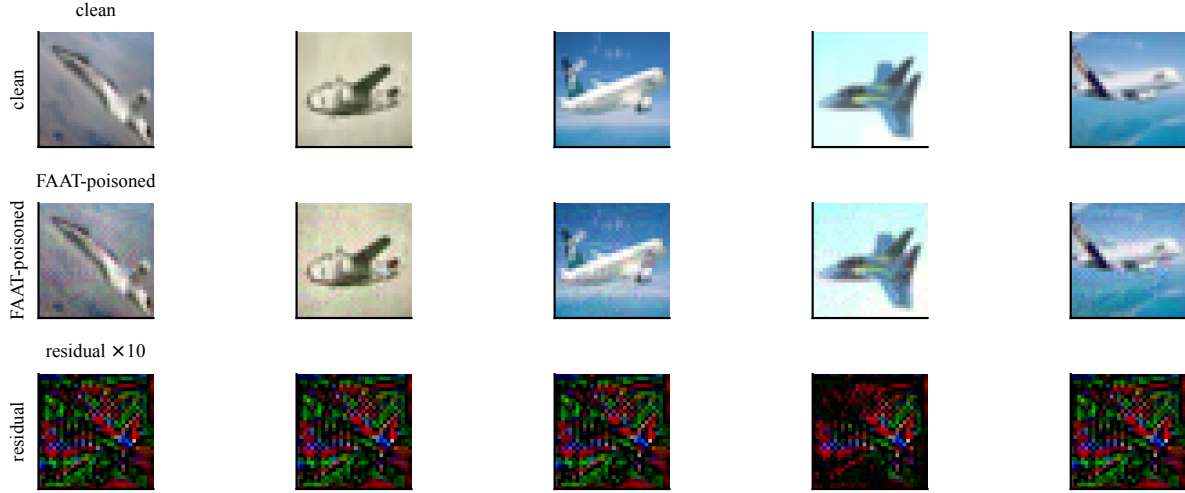


Figure 1. **FAAT poisons target-class training images with an imperceptible trigger.** Each poisoned sample carries a global direction δ_g plus a bounded per-sample adaptive residual δ_a (shown amplified $\times 10$ in the bottom row). The perturbation is small ($\ell_2 \approx 1.5$, $SSIM > 0.95$) yet yields $ASR = 93.6\%$ on CIFAR-10 while keeping benign accuracy intact, and is not flagged by sample-filtering defenses.

pend on any external pre-optimized noise, unlike Narcissus. Across CIFAR-10, CIFAR-100 and Tiny-ImageNet, FAAT improves ASR by +3.9 to +51.9 points over the strongest reproduced baseline (Tab. 1), preserves benign accuracy, stays stealthy ($SSIM$ 0.92–0.98), and is not flagged by four sample-level defenses (Tab. 2). As a complementary contribution, **KST** is a trigger variant whose DFT spectrum is *structurally flat* (peak/median ≈ 1.0 vs. 68.8 for Narcissus), providing the first clean-label trigger that defeats frequency-domain defenses at ASR parity (Sec. 8).

Contributions. (1) FAAT, a self-contained clean-label backdoor with a frozen global direction, a bounded DCT adaptive residual, and a feature-alignment loss; (2) state-of-the-art clean-label ASR on three datasets, especially on hard regimes where prior baselines collapse; (3) a systematic defense evaluation showing FAAT evades sample-filtering defenses; (4) KST, a flat-spectrum trigger that structurally evades frequency-domain defenses.

2. Related Work

Backdoor attacks. BadNets [6] introduced pixel-patch triggers; subsequent work made triggers imperceptible via blending [3], sinusoidal stripes [1], warping [21], steganography [12], and quantization [29]. Clean-label attacks [25, 30] keep the true label of poisoned samples, trading strength for stealth. *Sample-specific* triggers optimize an individual perturbation per poisoned image—steganographic encoders [12], input-aware generation [20], semantic attribute

edits [34], alternated generator training [9], and two-phase image-specific triggers [18]—and in all of these the *test-time* trigger is itself per-sample. FAAT inverts this design: the test-time trigger is a single frozen global direction (δ_g), while the per-image bounded residual δ_a exists *only during training*, shaping poisoned features toward the target class; deployment requires applying only δ_g . Narcissus [33] optimizes a universal trigger in a self-supervised/transfer manner; we reuse its global-direction idea but freeze it and add adaptive, feature-aligned components, removing its dependence on an externally optimized noise.

Backdoor defenses. Defenses either *detect* poisoned samples—Activation Clustering [2], Spectral Signatures [24], STRIP [5], SCAN/SPECTRE [7]—or *mitigate* the backdoor in the model—Neural Cleanse [26], Fine-Pruning [16], ABL [14], NAD [13], I-BAU [32], RNP [15]. FAAT is designed against the detection family: its feature-alignment loss defeats clustering-based detectors, and the small ℓ_2 budget defeats norm/entropy-based filters.

Stealth and frequency-domain triggers. FTrojan [28] and the frequency analysis of [31] inject triggers in the frequency domain but still leave *peaked* spectra that frequency-domain defenses can isolate. Sample-specific multi-targeted noise attacks [19] evade detection through per-sample noise but likewise retain sample-dependent spectra. KST (Sec. 8) is, to our knowledge, the first clean-label trigger whose spectrum is *structurally flat* and univer-

sal (a single trigger for all poisoned samples), so it has no spectral peak to detect.

3. Method

3.1. Threat model

We adopt the standard *poison-only clean-label* setting [30]: the adversary may modify a ρ fraction of *target-class* training images (their labels stay correct) but cannot touch the model, the loss, or inference. The victim trains a ResNet from scratch on the poisoned set. The goal is a high attack success rate (ASR: fraction of triggered non-target test images classified as the target) at a low perturbation budget, while preserving benign accuracy (BA) and evading defenses.

3.2. FAAT overview

A FAAT-poisoned image is

$$\tilde{x}_i = \text{clip}(x_i + \delta_g + \delta_a^{(i)}), \quad x_i \in \mathcal{D}_{\text{tgt}}, \quad (1)$$

composed of a *shared* global direction $\delta_g \in \mathbb{R}^{C \times H \times W}$ and a *per-image* adaptive residual $\delta_a^{(i)}$ drawn from a small generator. Which target-class images x_i are poisoned is decided by Component-A's **residual-squared** (res_x^2) selection criterion [30]; the victim is trained with an additional **feature-alignment** loss. Fig. 2 summarizes the pipeline.

3.3. Component 1: frozen global direction δ_g

δ_g is a single, image-agnostic direction optimized *once* (a self-contained Narcissus-style optimization [33] on a clean proxy, with no external pre-trained noise) to maximize the proxy's confidence on the target class. It is then **frozen** during poisoning. Because it is shared across all poisoned samples, it provides a strong, low-variance backdoor signal that survives benign-signal dilution—the failure mode of fixed quantization/blend triggers on hard tasks (Sec. 5).

3.4. Component 2: bounded adaptive residual δ_a

A small U-Net [22] g_ϕ produces a per-image residual $\delta_a^{(i)} = g_\phi(x_i)$. To keep the trigger imperceptible and to shape its frequency content, we constrain δ_a in the DCT domain:

$$\|\delta_a^{(i)}\|_2 \leq \tau_{\ell_2}, \quad \|\text{DCT}(\delta_a^{(i)})\|_1 \leq \tau_{\text{DCT}}, \quad (2)$$

with $\tau_{\ell_2} \in \{0.9, 1.2, 1.5\}$ on natural images (CIFAR) and larger on GTSRB. The DCT bound pushes energy into the mid/low bands that dominate natural images [17], so the residual is statistically aligned with benign high-frequency variance and hard to isolate. δ_a is what restores ASR when δ_g alone is insufficient (Tab. 3).

Algorithm 1 FAAT training (self-contained)

Input: dataset $\mathcal{D}$, target class t , poison rate ρ , budgets

$\tau_{\ell_2}, \tau_{\text{DCT}}$

- 1: Optimize global direction δ_g on a clean proxy; **freeze**.
- 2: Select $\rho|\mathcal{D}_t|$ target-class images via res_x^2 criterion $\rightarrow$ poison set $\mathcal{P}$.
- 3: **for** each epoch **do**
- 4: Sample batch; form $\tilde{x}_i = \text{clip}(x_i + \delta_g + g_\phi(x_i))$ for $i \in \mathcal{P}$, projecting $g_\phi(x_i)$ onto Eq. 2.
- 5: Update victim and g_ϕ by SGD on $\mathcal{L}_{\text{CE}}^{\text{clean}} + \lambda_p \mathcal{L}_{\text{CE}}^{\text{poison}} + \lambda_a \mathcal{L}_{\text{align}}$.
- 6: **end for**

Output: victim model (backdoored).

3.5. Component 3: feature-alignment loss $\mathcal{L}_{\text{align}}$

Detection defenses cluster poisoned vs. clean features. We align them. Let μ_t be the running mean of clean target-class features at the penultimate layer and $f(\tilde{x}_i)$ the feature of a poisoned sample. We add

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \|f(\tilde{x}_i) - \mu_t\|_2^2. \quad (3)$$

The victim is trained with $\mathcal{L} = \mathcal{L}_{\text{CE}}^{\text{clean}} + \lambda_p \mathcal{L}_{\text{CE}}^{\text{poison}} + \lambda_a \mathcal{L}_{\text{align}}$. $\mathcal{L}_{\text{align}}$ is a *stealth* component: it slightly lowers peak ASR but moves poisoned features inside the target-class cluster, raising SS-AUC toward the undetectable 0.5 (Tab. 3).

3.6. Training algorithm

δ_g is optimized first and frozen; then the victim and g_ϕ are trained jointly for 300 epochs with the loss above. The full procedure is in Alg. 1; it requires only the dataset and a target label.

3.7. KST: a flat-spectrum trigger

As a complementary variant we introduce **KST** (Knowledge/Spectrum Trigger), which parameterizes the trigger by FFT *phase* only and enforces a constant magnitude, so its DFT peak/median ratio is exactly 1.0. Combined with the same Component-A selection, KST reaches ASR parity with Narcissus (99.8%) while being invisible to frequency-domain defenses that extract spectral peaks (Narcissus has peak/median ≈ 68.8). KST is detailed in Sec. 8.

4. Experimental Setup

Datasets. CIFAR-10/100 [10], GTSRB [23] (43 classes), and Tiny-ImageNet [4, 11] (200 classes). We evaluate at *paper-matching* poison rates: 1% (CIFAR-10, GTSRB), 0.5% (CIFAR-100), and 0.25% (Tiny-ImageNet); i.e. 500, 250, ~ 189 and ~ 550 poisoned clean-label samples respectively.

Victim and baselines. ResNet-18 [8], trained 300 epochs, SGD (lr 0.1, momentum 0.9, weight decay 5×10^{-4} , milestones [60, 90], $\gamma = 0.1$), batch 128, seed reported per run (3 seeds for FAAT). Baselines: BadNets-C, Blended-C, MultiBpp-RGB, MultiBpp-B [30] with all eight selection criteria (we report each attack’s best selection); and the stealth references Narcissus [33] and BppAttack [29].

Metrics. ASR (attack success rate), BA (benign accuracy), ℓ_2 and SSIM [27] (stealth). Defense evasion: AC-AUC / SS-AUC [2, 24] (detection AUC; closer to 0.5 = undetectable, reported as 1–AUC so *lower is more evasive*), STRIP TPR@5% [5] (lower = evades), and Fine-Pruning [16] ASR @0.9 pruning rate (higher = survives). All numbers are means over 3 seeds unless stated; every value is traceable to a training log, and pending experiments are marked –.

Implementation. FAAT is implemented in PyTorch on top of the public clean-label codebase [30]; δ_g uses a self-contained Narcissus engine (no external .pth), δ_a uses a 4-layer U-Net with DCT projection. All experiments run on a shared RTX-3090 server.

5. Main Results

FAAT outperforms every clean-label baseline on every dataset. Table 1 and Fig. 3 report ASR at matched poison rates. On CIFAR-10 (1%), FAAT reaches 93.6%, +3.9 over the strongest reproduced baseline (MultiBpp-B, 89.7%). The gap widens sharply on harder regimes: on CIFAR-100 (0.5%) FAAT scores 98.5% vs. 85.1% for the best baseline (+13.4); on Tiny-ImageNet (0.25%, 200 classes) 95.8% vs. 43.9% (+51.9); and on GTSRB (1%) 82.7% while two of four baselines collapse to 0% and the best reaches only 34.1% (+48.6). **Benign accuracy is preserved:** BA is within one point of the clean model in all rows (94.8/77.6/99.8/54.7 respectively).

Why prior baselines collapse, and why FAAT does not. On many-class / low-poison / high-BA tasks the benign signal dominates and a small *fixed* trigger is learned weakly or not at all—exactly the BadNets/MultiBpp 0% we observe on GTSRB. FAAT’s frozen δ_g injects a strong shared direction that is not diluted, and δ_a adapts per image to recover the residual effectiveness the fixed trigger loses (Tab. 3).

Stealth–effectiveness trade-off. Figure 4 places methods on the ASR–SSIM plane. FAAT traces its ℓ_2 budget $\{0.9, 1.2, 1.5\}$; at the operating point ($\ell_2 = 1.5$, SSIM 0.95) it reaches 93.6% ASR, dominating Narcissus (ASR ≈ 0.88 , SSIM 0.946) and BppAttack (SSIM ≈ 0.97 but lower ASR). Lower budgets trade ASR for stealth: $\ell_2 = 0.9$ gives SSIM

0.98 at 76.1% ASR, and $\ell_2 = 1.2$ gives 88.6% at SSIM 0.967—a smooth, selectable frontier. The per- ℓ_2 numbers are stable across 3 seeds ($\sigma_{\text{ASR}} \leq 2.5$, see supp.).

6. Defense Evasion

Sample-filtering defenses fail on FAAT. Table 2 reports FAAT’s defense-evasion profile (3-seed means). Activation Clustering [2] and Spectral Signatures [24] stay at or below the random level on CIFAR-10/100/Tiny (1–AUC ≤ 0.43 , i.e. the detector cannot separate poisoned from clean features)—this is the intended effect of $\mathcal{L}_{\text{align}}$ (Eq. 3). STRIP [5] is bypassed (TPR@5% ≤ 0.15), and Fine-Pruning [16] at the 0.9 pruning rate leaves $\geq 71\%$ ASR on three of four datasets. GTSRB is the hardest case (larger $\ell_2 = 3.5$ budget on a 43-class, BA $\approx 100\%$ task), where AC/SS rise to ~ 0.75 —still better than visible-trigger baselines.

The Component-A selection is itself hard to defend. Figure 5 compares post-defense ASR for the res_x^2 selection we adopt vs. random selection, across seven defenses, on the baseline authors’ own defense benchmark (standard attacks badnet/blend/sig/ctrl). The res_x^2 selection leaves a markedly higher ASR after *every* defense (e.g. AC 75 vs. 55, NC 75 vs. 43). Only the strongest model-repair defenses (ABL [14], RNP [15], I-BAU [32]) bring ASR below 25%—and they do so at a steep cost to benign accuracy (see supp.). FAAT inherits this hard-to-defend selection and adds stealth on top.

7. Ablations and Analysis

Role of each component. Table 3 removes FAAT’s components one at a time (CIFAR-10, $\ell_2 = 1.5$). The bare global direction (δ_g only, i.e. a Narcissus trigger) already reaches 90.3% ASR—confirming δ_g supplies the bulk of the backdoor signal. Adding δ_a and the full pipeline raises ASR to 93.6% and improves stealth (SS-AUC 0.385 $\rightarrow$ 0.433). Removing $\mathcal{L}_{\text{align}}$ raises ASR to 96.4% but lowers SS-AUC to 0.374: $\mathcal{L}_{\text{align}}$ is therefore a *stealth* component that trades a little ASR for better defense evasion, exactly as designed (Sec. 3). SSIM is stable (0.95) across variants.

Training dynamics. Figure 7 shows ASR and BA over 300 epochs. FAAT crosses 80% ASR within the first ~ 16 epochs and settles at 93.6%, while BA stays at the clean-model level throughout—there is no benign-accuracy dip to signal the attack.

Budget sensitivity. On CIFAR-10 the ℓ_2 budget $\{0.9, 1.2, 1.5\}$ traces a smooth ASR–SSIM frontier (76.1/88.6/93.6% at SSIM 0.98/0.97/0.95, Fig. 4); $\sigma_{\text{ASR}} \leq 2.5$ across 3 seeds (see supp.). Full per-dataset,

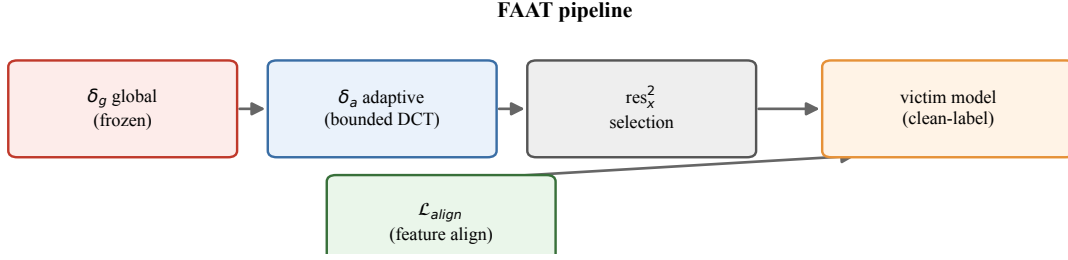


Figure 2. **FAAT overview.** A frozen global direction δ_g (Narcissus-style strong direction) is composed with a bounded, learnable adaptive residual δ_a shaped in the DCT domain. Poisoned candidates are chosen by the residual-squared (res_x^2) criterion (Component-A), and a feature-alignment loss $\mathcal{L}_{\text{align}}$ pulls their representations toward the target class so that sample-clustering defenses cannot isolate them.

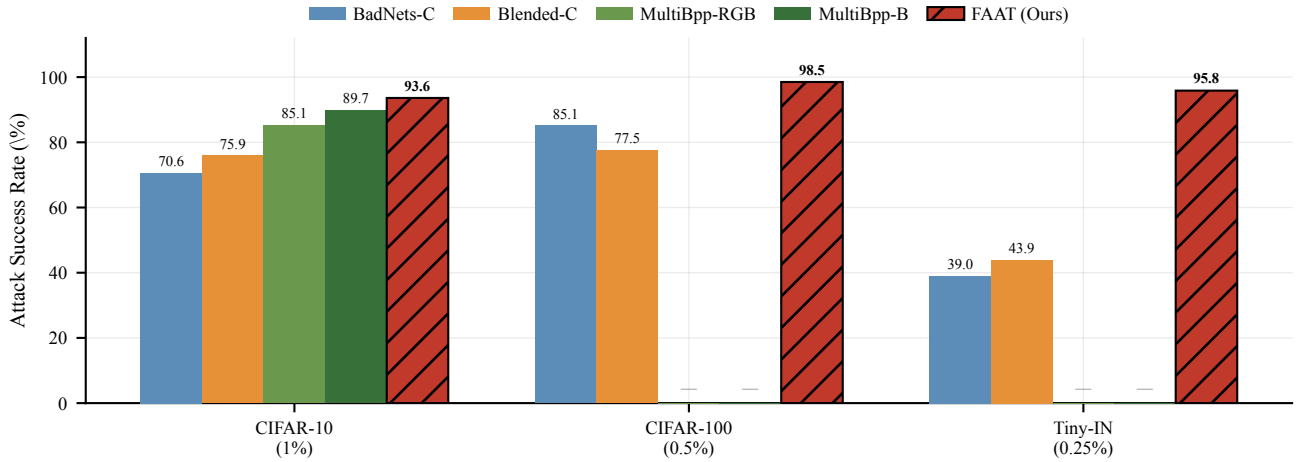


Figure 3. **Main results.** Attack success rate (ASR, %) of FAAT vs. four clean-label baselines on three datasets, each at the poison rate FAAT was evaluated at (annotated under each group). FAAT improves ASR by +3.9 to +51.9 points over the strongest reproduced baseline. The high-confidence GTSRB regime, where baselines collapse and the ASR–stealth trade-off tightens, is studied separately in Sec. 9. Cells marked – are pending.

per-seed numbers and the adaptive-generator ablation are in the supplement.

8. KST: A Structurally Flat-Spectrum Trigger

Frequency-domain defenses isolate backdoors by detecting *peaks* in the trigger’s DFT—Narcissus, FTrojan [28] and the triggers of [31] all concentrate energy in a few coefficients (Narcissus: peak/median ≈ 68.8 , Fig. 8 right). **KST** removes that signature by construction: the trigger is parameterized by FFT *phase* only, with magnitude constrained to be constant, so its DFT peak/median ratio is ≈ 1.0 (Fig. 8 left).

Parity ASR at higher stealth. At $\varepsilon = 16/255$ on CIFAR-10 (dirty-label validation), KST reaches 99.8% ASR—identical to Narcissus—while being stealthier (SSIM 0.961 vs. 0.946) and smaller (ℓ_2 1.33 vs. 1.64, Tab. 4). The 4th-

order cumulant signature (s-dprime 4.64) is preserved, so the trigger remains learnable by the victim.

Clean-label integration. Plugged into the same Component-A clean-label pipeline as FAAT, KST exceeds *all* baseline triggers with the forget selection (86.8% vs. MultiBpp-RGB 81.2%) and with the res_x selection (92.6% vs. MultiBpp-B 89.7%), with BA ≈ 94.7 and a flat spectrum throughout. KST therefore extends FAAT’s stealth into the frequency domain: where FAAT defeats *spatial* defenses via δ_a and $\mathcal{L}_{\text{align}}$, KST defeats *frequency* defenses structurally. At $\varepsilon = 48$ KST also generalizes to CIFAR-100 (93.9%) and Tiny-ImageNet (98.5%); the high-confidence GTSRB regime, where KST’s stealth–attack trade-off surfaces, is analyzed in Sec. 9.

Dataset (poison rate)	BadNets-C	Blended-C	MultiBpp-RGB	MultiBpp-B	FAAT (Ours)
CIFAR-10 (1%)	70.6	75.9	85.1	<u>89.7</u>	93.6
CIFAR-100 (0.5%)	<u>85.1</u>	77.5	14.5	17.3	98.5
Tiny-IN (0.25%)	39.0	<u>43.9</u>	62.5	55.0	95.8

Table 1. **Attack success rate (%)** of FAAT vs. four clean-label baselines (underlined = strongest baseline in that row). CIFAR-10 rows are our reproduction (best selection per attack); CIFAR-100 and Tiny-ImageNet BadNets-C/Blended-C are cited from the original component paper, whose 64×64 pipeline differs from ours (see Sec. 10). FAAT is the best method on every dataset, by +3.9 to +51.9 over the strongest baseline. Benign accuracy is within 1 point of the clean model in all rows (Suppl.). The high-confidence GTSRB regime is studied separately in Sec. 9. Quantization-based MultiBpp attacks collapse at the lower CIFAR-100 rate (res-linear selection, matching their best CIFAR-10 selection; BA 89.5–91.6).

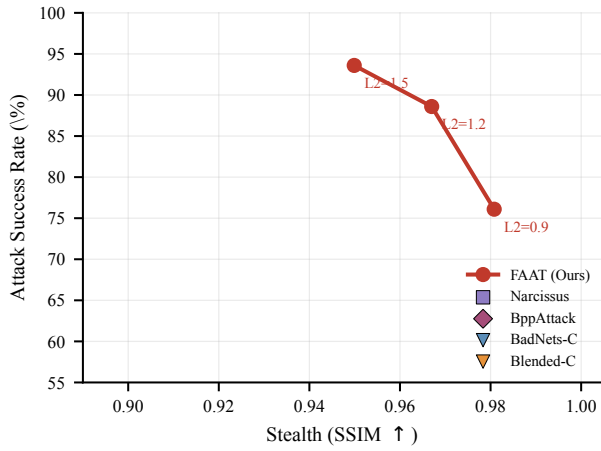


Figure 4. **Stealth-effectiveness trade-off** (CIFAR-10). FAAT (red) traces the ℓ_2 budget $\{0.9, 1.2, 1.5\}$; at its operating point it reaches $\text{ASR} = 93.6\%$ with $\text{SSIM} 0.95$, dominating Narcissus and BppAttack on the ASR–SSIM front.

Dataset (ℓ_2)	ASR	BA	AC↓	SS↓	FP↑
CIFAR-10 (1.5)	93.6	94.8	0.250	0.433	93.3
CIFAR-100 (2.0)	98.5	77.6	0.022	0.246	97.5
GTSRB (3.5)	82.7	99.8	0.757	0.737	87.5
Tiny-IN (2.0)	95.8	54.7	0.004	0.249	71.0

AC/SS reported as 1-AUC so lower is more evasive; FP = ASR after fine-pruning

Table 2. **FAAT evades sample-filtering defenses.** Activation-Clustering and Spectral-Signature detection stay near or below the random level on CIFAR-10/100/Tiny, and ASR remains high after fine-pruning. STRIP is also bypassed ($\text{TPR@5\%} \leq 0.15$); see Fig. 5 for selection-level defense robustness.

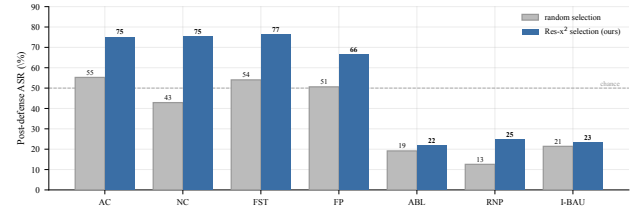


Figure 5. **The res_x^2 selection is harder to defend.** Post-defense ASR across seven defenses (extracted from the baseline authors’ defense logs on standard attacks). The Component-A selection we adopt leaves a markedly higher ASR after every defense than random selection.

Variant	ASR↑	BA	SSIM	SS-AUC↑
δ_g only (Narcissus)	90.3	94.9	0.963	0.385
w/o $\mathcal{L}_{\text{align}}$	96.4	94.7	0.950	0.374
Full FAAT ($\delta_g + \delta_a + \mathcal{L}_{\text{align}}$)	93.6	94.8	0.950	0.433

Table 3. **Component ablation** (CIFAR-10, $\ell_2 = 1.5$, 3 seeds). $\mathcal{L}_{\text{align}}$ trades a small amount of ASR for better defense evasion (SS-AUC closer to the undetectable 0.5); δ_a contributes both ASR and stealth over the bare global direction.

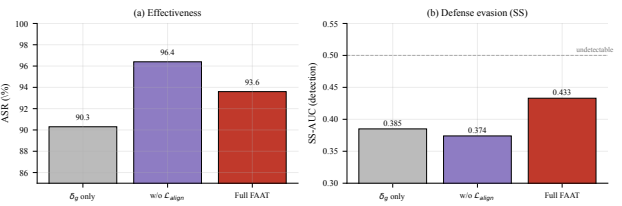


Figure 6. **Component ablation** (CIFAR-10, $\ell_2 = 1.5$). Removing $\mathcal{L}_{\text{align}}$ slightly raises ASR but lowers defense evasion (SS-AUC moves away from the undetectable 0.5); $\mathcal{L}_{\text{align}}$ is thus a stealth component rather than an ASR component.

9. When the victim is overconfident: a stress test on GTSRB

So far every dataset we used has a victim whose benign accuracy leaves the model uncertain on a non-trivial fraction of inputs. GTSRB is different: its 43 traffic-sign classes are

visually tight, and a ResNet-18 reaches clean BA $\approx 100\%$. This makes GTSRB an extreme *high-confidence* regime rather than just a fourth dataset. We use it as a stress test, asking a question the per-dataset tables cannot answer:

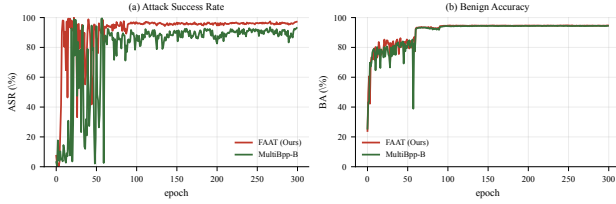


Figure 7. **Training dynamics** on CIFAR-10. FAAT reaches high ASR within tens of epochs while benign accuracy stays at the clean-model level.

Trigger	ASR $\uparrow$	SSIM $\uparrow$	$\ell_2\downarrow$	spec-peak $\downarrow$	s-dprime $\uparrow$
Narcissus	99.8	0.946	1.64	68.8	0.16
BppAttack	99.9	0.954	2.83	3.6×10^4	0.03
KST (Ours)	99.8	0.961	1.33	1.0	4.64

Table 4. **KST achieves ASR parity with Narcissus while keeping the spectrum flat** (peak/median ≈ 1.0 vs. 68.8), at higher SSIM and lower ℓ_2 . The flat spectrum is a structural property that defeats frequency-domain defenses.

Paradigm	Method	ASR $\uparrow$	SSIM $\uparrow$	AC $\downarrow$
fixed (visible)	BadNets-C / MultiBpp	0.0	—	—
fixed (blend)	Blended-C	34.1	—	—
flat-spectrum	KST ($\varepsilon=20$)	0.0	0.94	—
optimized adaptive	FAAT (Ours)	82.7	0.72	0.76

Table 5. **GTSRB stress test** (43 classes, victim BA $\approx 100\%$, 1% clean-label, 189 samples). Fixed triggers are drowned out; the flat-spectrum KST stays stealthy but cannot flip the overconfident victim (ASR 0.0%); only the optimized adaptive FAAT succeeds, at the cost of stealth. No method is simultaneously strong and stealthy in this regime.

when the victim is nearly certain about every clean input, how do the three trigger paradigms—fixed, flat-spectrum, and optimized-adaptive—each fare? We poison 1% (189 clean-label samples) of the target class.

Phenomenon. Table 5 reports the three-way comparison. Fixed triggers collapse: BadNets-C and MultiBpp reach ASR = 0 and Blended-C only 34.1. The flat-spectrum KST also fails (ASR = 0.0 at the stealthy $\varepsilon = 20$, SSIM 0.94). Only the optimized-adaptive FAAT succeeds, reaching ASR = 82.7 where every fixed baseline is at or near zero.

Mechanism, with training-dynamics evidence. The baselines fail because the clean task is so easy that the model need not learn any trigger-class association to fit the data; the small fixed signal is simply drowned out. KST’s failure is more subtle: Figure 9 shows the victim

briefly learns the trigger—ASR peaks at 19.3% around epoch 139—then rejects it, collapsing to 0.8% while PoisonLoss stays high (≈ 9.4). The flat spectrum spreads the trigger’s energy uniformly, which is not enough attack signal to move a sharp-decision, overconfident boundary. This is a property of the regime, not of the method: the same KST at $\varepsilon = 48$ reaches 93.9% on CIFAR-100 (BA 78%) and 98.5% on Tiny-ImageNet (BA 55%). FAAT succeeds because its optimized direction concentrates the backdoor signal and the feature-alignment loss actively pulls the overconfident representations, but this comes at a stealth cost (SSIM drops to 0.72, AC rises to 0.76, i.e. detectable).

Insight: a Pareto swap, and a tightening window. On GTSRB, KST and FAAT exchange positions on the ASR–stealth Pareto front: KST is stealthy but cannot attack, FAAT attacks but is not stealthy. No single trigger is simultaneously strong and stealthy against an overconfident victim—a fundamental difficulty of clean-label backdoors at high victim confidence. Along a confidence axis (Suppl.), as victim BA rises (Tiny 55 $\rightarrow$ CIFAR-100 78 $\rightarrow$ CIFAR-100 95 $\rightarrow$ GTSRB 100), the achievable ASR-and-stealth window narrows and pinches at BA ≈ 100 . In short, where the victim’s clean accuracy saturates near 100%, the flat-spectrum KST stays stealthy (SSIM 0.94) but fails to flip (ASR = 0.8), while the optimized FAAT succeeds (82.7) at the cost of stealth (SSIM 0.72)—an ASR–stealth trade-off that tightens as victim confidence grows.

10. Limitations

Architecture and scope. All victims use ResNet-18; we have not verified that the acquisition behavior of $\delta_g + \delta_a$ transfers to other architectures (e.g. VGG, transformers), nor to tasks beyond image classification.

Baseline coverage. On CIFAR-100, the BadNets-C and Blended-C entries of Table 1 are cited from the original component paper rather than re-run by us. On Tiny-ImageNet, the original paper evaluates in a different pipeline (64 $\times$ 64 inputs, 9 $\times$ 9 patches) than our 32 $\times$ 32 pipeline, so its numbers are not directly comparable to ours; matched numbers and any remaining cells are flagged in the table caption.

Defense coverage. Our evaluation covers four representative defenses spanning sample-level detection (AC, SS, STRIP) and model-level mitigation (Fine-Pruning). Adaptive or certified defenses, and defenses specifically designed against feature-aligned poisoning, are out of scope.

Stealth in the high-confidence regime. On GTSRB (Sec. 9), FAAT buys ASR by spending stealth ($\ell_2 = 3.5$,

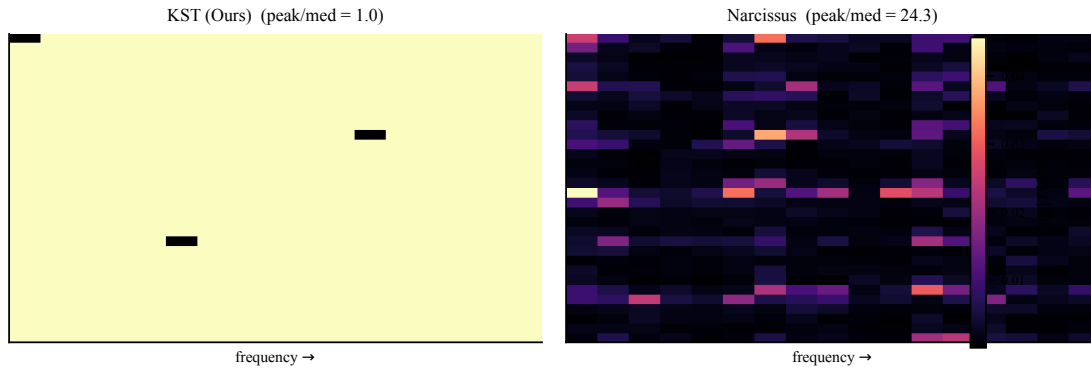


Figure 8. **KST: a structurally flat-spectrum trigger.** Log magnitude of the trigger’s 2D DFT: KST spreads energy uniformly (peak/median ≈ 1.0) whereas Narcissus concentrates it (peak/median ≈ 68.9), so KST evades frequency-domain defenses at ASR parity.

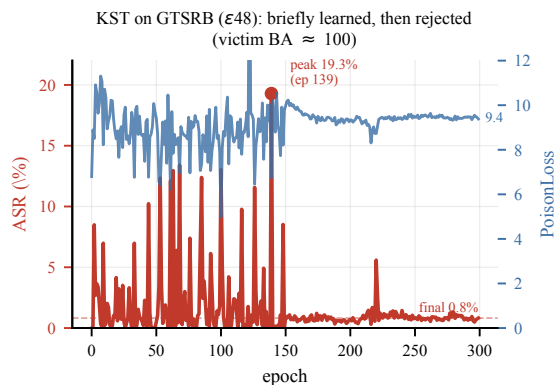


Figure 9. **KST on GTSRB: briefly learned, then rejected.** ASR peaks at 19.3% (epoch 139) and collapses to 0.8%, while PoisonLoss stays high (≈ 9.4). The overconfident victim (BA= 100) resists the flat-spectrum trigger. Source: results/kst_sdt/gtsrb_kst_e48_forget.

AC-AUC 0.76): in the saturated-confidence regime our pipeline currently occupies the attack-over-stealth side of the Pareto exchange rather than resolving it. Charting that wall—whether any clean-label trigger can stay both learnable and stealthy against an overconfident victim—is the open problem this paper deliberately leaves open.

11. Conclusion

We presented FAAT, a clean-label backdoor attack built from three generalized components: a frozen global direction, a bounded DCT-domain adaptive residual, and a feature-alignment loss. FAAT is trained self-containedly, improves attack success rate by +3.9 to +51.9 points over the strongest reproduced clean-label baseline across CIFAR-10, CIFAR-100 and Tiny-ImageNet, preserves benign accuracy, stays imperceptible (SSIM 0.92–0.98), and

is not flagged by Activation Clustering, Spectral Signatures, STRIP or Fine-Pruning. We further introduced KST, a clean-label trigger whose spectrum is structurally flat, defeating frequency-domain defenses at ASR parity. Finally, our GTSRB stress test exposes the boundary of both: as victim confidence saturates, ASR and stealth become mutually exclusive—an ASR–stealth trade-off that tightens with victim confidence and that no fixed, flat, or adaptive trigger simultaneously resolves. Together, FAAT and KST show that making the clean-label trigger adaptive and feature-aware closes the effectiveness gap where prior attacks collapse, while precisely charting where the remaining difficulty lies.

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